Erdős #1093: the deficiency of $\binom{n}{k}$ is defined on a domination locus, and is decidable for each k

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Abstract

The deficiency of $\binom{n}{k}$ is defined only when no prime $p \leq k$ divides it, and equals $\#\{0 \leq i < k : \ n-i$ is k-smooth$\}$. We prove that the definedness hypothesis is exactly the digit-domination criterion whose least solution is $g(k)$ in #1095, so the two problems share one locus; and we observe that combined with the Erdős–Lacampagne– Selfridge bound this makes the deficiency-> 1 question *decidable for each fixed k*. #1093 is not proved.

1 The definedness locus

Write $k \preceq_p n$ for base-p digit domination.

Proposition 1. *The deficiency of $\binom{n}{k}$ is defined if and only if $k \preceq_p n$ for every prime $p \leq k$. This is precisely the condition whose least solution $n > k+1$ is $g(k)$ in #1095.*

Proof. Definedness means no prime $p \leq k$ divides $\binom{n}{k}$. By Kummer, $p \mid \binom{n}{k}$ iff the base-p addition $k + (n-k)$ carries, iff some base-p digit of k exceeds that of n. ☐

So #1093 and #1095 are one condition asked twice: #1095 for the least point of the locus, #1093 for the smoothness of the k consecutive integers $n-k+1, \ldots, n$ at points of it.

Corollary 2 (decidability for fixed k)**.** *Erdős, Lacampagne and Selfridge proved that if the deficiency exists and is ≥ 1 then $n \ll 2^k\sqrt{k}$. Combined with Proposition 1, for each fixed k the set of n with deficiency ≥ 1 lies in an explicit finite range and membership is decided by digit domination together with k-smoothness of k consecutive integers. Hence* "are there only finitely many binomial coefficients of deficiency > 1?" *is decidable separately for each k, and the open content of #1093 is uniformity in k.*

This is a materially softer shape than the cross-base intersection problems in the same cluster (notably #377), where no finite range suffices.

2 Structure of the deficiency

Write $P^+(m)$ for the largest prime factor of m, and $D(n,k)$ for the deficiency of $\binom{n}{k}$ when it is defined.

Theorem 3. *Suppose $D(n,k)$ is defined. Then*

(a) $\gcd\big(\binom{n}{k},\ k!\big) = 1$*;*

(b) $D(n,k) = \#\big\{m \in (n-k,\,n] \ :\ P^{+}(m) \le k\big\}$*, the count of k-smooth integers in a window of length k ending at n;*

(c) *if $D(n+1,k)$ is also defined, then $\big|D(n+1,k) - D(n,k)\big| \le 1$.*

Proof. (a) Definedness says no prime $p \le k$ divides $\binom{n}{k}$. Every prime factor of $k!$ is at most k, so $\binom{n}{k}$ and $k!$ share no prime factor.

(b) By definition $D(n,k) = \#\{0 \le i < k : \ n-i \text{ is } k\text{-smooth}\}$, and $m = n-i$ runs over exactly the integers of $(n-k,n]$ as i runs over $\{0,\dots,k-1\}$; k-smoothness of m is the statement $P^{+}(m) \le k$.

(c) By (b), $D(n+1,k)$ counts the k-smooth integers of $(n+1-k,\,n+1]$. This window is obtained from $(n-k,\,n]$ by deleting the single integer $n-k+1$ and adjoining the single integer $n+1$. Deleting one element changes the count by at most 1 and adjoining one changes it by at most 1, in opposite directions, so the two counts differ by at most 1. □

Verified: (a) on all defined pairs with $3 \le k < 10$, $n < 600$; (b) on all defined pairs with $3 \le k < 13$, $n < 20000$; (c) on all 2092 pairs of consecutive defined n in the same range.

Corollary 4. $D(n,k) \le \Psi_k := \max_x \#\{m \in (x-k,x] : P^{+}(m) \le k\}$*, the largest number of k-smooth integers in any window of length k. In particular the question of whether deficiency > 1 occurs infinitely often is subsumed by the question of how often two k-smooth integers lie within k of one another.*

Proof. Immediate from Theorem 3(b). □

Measured: in the census below, the deficiency-> 1 examples all exhibit exactly this coincidence — at $k = 10$ both $n = 46$ and $n = 47$ realise $D = 3$ on the same smooth triple $\{40,42,45\}$, and at $k = 8$ the pair $\{40,42\}$ gives $D = 2$ at $n = 44$. The largest observed deficiency is 4, at $\binom{47}{11}$.

3 Census

Over $n < 40000$, $3 \le k < 14$:

- **Deficiency** 1: 25 examples, beginning $(n,k) = (7,3), (13,4), (14,4), (23,5), (62,6), (143,7), (89,8), (143,8)$ The first five reproduce the published list $\binom{7}{3}, \binom{13}{4}, \binom{14}{4}, \binom{23}{5}, \binom{62}{6}$ exactly; $(62,6)$ is again Selfridge's binomial.
- **Deficiency** > 1: 6 examples — $(44,8;d{=}2)$, $(46,10;3)$, $(47,10;3)$, $(74,10;2)$, $(47,11;4)$, $(174,12;2)$. The largest observed deficiency is 4, at $\binom{47}{11}$.

Both counts are still growing at the edge of the range, so neither question is settled by the data; but deficiency-> 1 examples are markedly rarer and cluster at small k, consistent with the conjectured finiteness.

Remark 5 (register)**.** ***Proved:*** *Proposition 1 (elementary from Kummer; not claimed new), Corollary 2, which combines it with the cited Erdős–Lacampagne–Selfridge bound, and Theorem 3 with Corollary 4, which recast the deficiency as a count of k-smooth integers in a window of length k and give its one-step Lipschitz property.* ***Measured:*** *the two censuses, and the agreement of the first five deficiency-*1 *examples with the published list.* ***Not proved:*** *either the infinitude of deficiency* 1 *or the finiteness of deficiency* > 1*.* ***Falsifier:*** *a deficiency-*> 1 *example with $k \geq 20$ would weaken the finiteness conjecture; none was found here.* ***Next:*** *exhaust each $k \leq 20$ over the ELS93 range $n \ll 2^k\sqrt{k}$, which by Corollary 2 settles deficiency* > 1 *for those k outright.*